

A. RESEARCH PROFILE

Research scientist at **Qwen**, responsible for the subjective capabilities of frontier models since Qwen 3.6: how to make sustained human–AI interaction feel genuinely useful, adaptive, and trustworthy. Originates and leads end-to-end research programs spanning mechanism discovery and optimization theory, production-scale inference systems, human-centered evaluation, and long-horizon simulation. Research agenda: **frontier language models, multi-agent and interactive AI, and AI evaluation**, toward persistent digital life and experiential world-game environments.

Previously a researcher in Stanford University’s NLP Group, advised by **Christopher D. Manning**; completed a B.S. at the University of Wisconsin–Madison, advised by **Sharon Li**. Lead author of **NeurIPS 2025 Spotlight MetaMind**, *Confident Decoding*, **ACL 2026 Cognition-of-Thought**, and **ICLR 2026 LH-Deception**.

B. SELECTED RESEARCH CONTRIBUTIONS

- 01

Test-time computation and reasoning systems

QWENLM RELEASE

Originated and led *Confident Decoding*, revealing Guess–Refine–Perturb dynamics in aligned language models and casting layer selection as an optimal-stopping problem. Built a training-free vLLM implementation validated across **3 model families and 6 dense/MoE backbones**: up to **+9.4 points** on LiveCodeBench, **zero memory overhead**, and **less than 2% latency overhead**.
- 02

Metacognitive and socially aligned intelligence

NEURIPS SPOTLIGHT / ACL 2026

Established and led a coherent program in machine interactive cognition: *MetaMind* models latent mental states through a norm-aware multi-agent metacognitive scaffold (**+35.7%** on real-world social scenarios; **+6.2%** on Theory-of-Mind reasoning), while *Cognition-of-Thought* turns alignment into an explicit, auditable rollback-and-refine loop. *Latent-RL* extends this agenda toward internalized mental-state reasoning.
- 03

Long-horizon human–AI interaction and world modeling

ICLR 2026

Originated and equal-first-authored *LH-Deception*, a multi-agent simulation and auditing framework evaluated across **11 frontier models**, establishing how event pressure produces chains of deception and erodes supervisor trust. Extended trajectory-level impact modeling to *MarioLM*’s interactive world architecture and a **233-participant randomized study** of AI-driven financial and emotional manipulation.

C. RESEARCH AND INDUSTRY EXPERIENCE

Feb 2026 – Present	Qwen Team, Alibaba Group Researcher A Star Top Talent Program. Research leads: Dr. An Yang and Tianyi Tang - Originated and led <i>Confident Decoding</i>, discovering Guess–Refine–Perturb dynamics and formulating entropy-guided layer selection as an optimal-stopping problem. Shipped a training-free vLLM implementation validated on 6 dense/MoE backbones: +6.5 GPQA-D points, 0 MB KV-cache, <1% FLOPs, and <2% latency. - Originated and led <i>MarioLM</i>, unifying interactive-world evaluation and optimization through hierarchically decomposed, causally grounded, and temporally continuous signals. <i>MarioEval</i> combines Scene/Latent-State/Event-Chain harnesses with L1–L6 cascade rubrics and user-interest halting; on 500+ annotated turns across four frontier models, it recovered distinct style, drift, persona, and agency bottlenecks (92.3–98.8% human agreement on L1–L4) and exposed LCGD and hedonic bias. <i>MarioOpt</i> reuses these traces for HRD curriculum rewards and EFPO trajectory recovery, grounding faithful user simulators and persistent experiential worlds. - Originated and led <i>Latent-RL</i>, a cognitive-internalization framework for distilling <i>MetaMind</i>’s external multi-agent Theory-of-Mind loop into native single-model reasoning. It factorizes policy over latent mental states and responses, couples a Forward/Inverse-MetaMind teacher–verifier duality with process rewards over belief, intent, emotion, moral constraints, and temporal consistency, and trains through MarioLM SFT → latent GRPO → latent compression. This System-2-to-System-1 program targets auditable training and efficient single-pass behavior across functional ToM, SOTOPIA, hallucination, and long-horizon consistency.
Jun 2025 – Dec 2025	Stanford University, NLP Group Visiting Student Researcher Advised by Prof. Christopher D. Manning - First-authored two NeurIPS 2025 works: <i>Dynamic Mode Steering</i> combines Information-Bottleneck theory, causal probing, and adaptive activation steering to shift LLM inference from memorization toward generalization (up to +6.4 points); <i>Deep Graph Learning</i> builds an interpretable graph-temporal policy simulator for industrial metrics prediction ($R^2 = 0.95$; 25% lower RMSE). - With Prof. Christopher D. Manning, framed human-centered AI as social reasoning under uncertainty: infer latent mental states from incomplete behavioral evidence, model causal interaction, and simulate long-horizon policy consequences—not merely scale data or optimize verifiable outputs. The agenda moves beyond language-only and hand-authored supervision through self-evolve and cognitive-process rewards, toward persistent, autonomous “experiential worlds”—metaverse-scale environments unifying simulation, evaluation, and training.

Apr 2025 – Dec 2025	Amazon AGI Advised by Dr. Rahul Gupta and Jwala Dhamala - Originated and equal-first-authored <i>LH-Deception</i>, the first controlled framework for quantifying deception across long-horizon, interdependent work. Designed performer–supervisor–auditor agent sandbox, a 14-task stream, and social-science-grounded pressure events to jointly trace deception and evolving trust. Across 11 frontier models, uncovered 21.4–79.3% deception rates, strong anti-correlations with supervisor trust ($r = -0.804$) and relational comfort ($r = -0.781$), 7.1% additional deceptive instances missed by per-event audits, and escalating chains of concealment and fabrication. - Collaborated on trustworthiness evaluation and optimization for Amazon Nova foundation models, translating behavioral risk analysis into evaluation design, failure diagnosis, and mitigation priorities for model development.	Student Researcher
Feb 2025 – Dec 2025	UW–Madison, Deep Learning Lab Advised by Prof. Sharon Li Originated and first-authored a cognitive-alignment program spanning NeurIPS 2025 Spotlight <i>MetaMind</i> and ACL 2026 <i>Cognition-of-Thought</i>. <i>MetaMind</i> recasts social understanding as staged metacognition—latent-state hypothesis generation, norm-aware refinement, social-memory conditioning, and self-validation—achieving SOTA across 16+ LLM backbones (+35.7% on real-world scenarios; +9.0% social cognition) and human-level ToM. <i>CooT</i> turns the same principle inward: a Generator–Perceiver decoder audits precedence-aware cognitive states, causally rolls back unsafe trajectories, and regenerates under adaptive guidance, reaching 0.80 AIR-Bench compliance (+13%) and +9.02% prosocial reasoning with auditable, policy-swappable inference-time alignment.	Researcher
Sep 2024 – Jan 2025	ByteDance Advised by Dr. Xingnan Wang - Designed and prototyped a unified multimodal GUI-agent architecture for iOS, spanning visual grounding, state tracking, action planning, and cross-application execution; contributed the resulting agent patterns to ByteDance’s Coze ecosystem. - Originated and led <i>Seeker</i>, the first systematic LLM framework for robust exception handling. Reconstructed expert practice as a Scanner–Detector–Predator–Ranker–Handler pipeline, grounded by a Common Exception Enumeration and Deep-RAG over a 433-node exception hierarchy. Achieved 91% sensitive-region coverage, 79% exception-type accuracy, and 92% code-review quality; improved SWE-bench exception-issue resolve/apply rates from 19/43% to 26/61% and CoderEval Pass@1 from 27.83% to 38.16%.	AI Architect Intern
Apr 2024 – Jan 2025	Zhipu AI & Tsinghua University, CoAI Lab Advised by Prof. Minlie Huang, Prof. Yuan Yuan - Co-led an LLM-based psychological-counseling simulation program for Emohaa, translating counseling protocols into multi-turn client simulation and counselor-response evaluation to improve empathetic, socially adaptive dialogue; integrated into Tsinghua University’s AI Emotion Companionship System. - Advanced a human-centered alignment agenda spanning causal human-subject experimentation and social-world evaluation. A randomized study of 233 participants (699 observations) showed that covertly misaligned AI advice increased the odds of selecting inferior options by 5.2–8.0× (up to +38 pp) while users still rated advisors helpful, with no reliable added effect from explicit manipulation tactics. <i>SocialEval</i> introduced bilingual World Trees and joint goal-achievement/interpersonal-ability evaluation across 153 social worlds and 2,493 probes, revealing 17.2–23.8-point human gaps in goal achievement, goal-defeating prosocial bias, and ability-specific neural partitions.	Research Scientist Intern
Jun 2023 – May 2024	Peking University, Key Lab of High Confidence Software Technologies, MoE Advised by Prof. Zhi Jin and Prof. Ge Li Co-developed a trustworthy code-intelligence evaluation stack spanning repository realism, benchmark freshness, and black-box contamination control. <i>DevEval</i> aligned 1,825 samples from 115 repositories with real code and dependency distributions, exposing a 53.04% GPT-4 Pass@1 ceiling; <i>EvoCodeBench</i> added a continuously refreshed collection pipeline, a 10-domain taxonomy, and DSI-based comfort/strange-domain analysis, reducing potential leakage from 41.47% to 2.18% and revealing only 20.74% GPT-4 Pass@1 on fresh repositories. <i>CDD/TED</i> detected explicit and implicit contamination from sampled outputs alone (+21.8–30.2% relative gains) and removed up to 66.9% of contamination-driven score inflation.	Researcher
Jan 2023 – Mar 2023	Beijing Zen-AI Technology Ltd. Advised by Libing Yan Developed production software for mission-critical, high-reliability systems in a regulated industrial environment; project scope, system architecture, and deployment details are withheld under NDA.	Software Engineer
Dec 2021 – Mar 2022	H3O Incubated in the XbotPark ecosystem. Advised by Prof. Zexiang Li, Prof. Bingqiang Gao, and Prof. Jie Gan Co-founded H3O and led 0-to-1 product strategy, system architecture, and prototyping for a connected low-water foot-care appliance. Translated 170+ user surveys, 20+ field/expert interviews, and 100+ product analyses into a modular system combining vortex cleaning, warm-air drying, replaceable care cartridges, sensorized safety controls, and app/wearable interaction; drove productization and venture strategy within XbotPark, helping secure \$500K seed financing.	Co-Founder and CTO

D. SELECTED PUBLICATIONS

Selected peer-reviewed and flagship research. Complete record: Google Scholar. *Equal contribution.

[1]

Deeper is Not Always Better: Mitigating the Alignment Tax via Confident Layer Decoding

Xuanming Zhang, Sining Zhoubian, Yuxuan Chen, Tianyi Tang, An Yang, Sean Du, Chujie Zheng, Fei Huang, Dayiheng Liu, Gao Huang, Jingren Zhou

Under review, *EMNLP 2026* **REVIEWER MEAN 3.67 / CURRENT TOP 1%**

Granted national patent; integrated into the official *QwenLM* repository.

[2]

MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems

Xuanming Zhang, Yuxuan Chen, Samuel Yeh, Sharon Li

Advances in Neural Information Processing Systems (NeurIPS), 2025 **SPOTLIGHT**

Recognized by Yoshua Bengio at *NeurIPS 2025 ResponsibleFM*; publicly highlighted by Lex Fridman; subsequently cited in a *Trends in Cognitive Sciences* perspective on AI metacognition co-authored by Bengio.

[3]

Cognition-of-Thought Elicits Social-Aligned Reasoning in Large Language Models

Xuanming Zhang, Yuxuan Chen, Samuel Yeh, Sharon Li

Annual Meeting of the Association for Computational Linguistics (ACL), 2026

[4]

Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions

Yang Xu*, Xuanming Zhang*, Samuel Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Sharon Li

International Conference on Learning Representations (ICLR), 2026

[5]

Human Preferences Are Susceptible to Covertly Misaligned AI Advice

Sahand Sabour, June M. Liu, Siyang Liu, Chris Z. Yao, Shiyao Cui, Xuanming Zhang, Wen Zhang, Yaru Cao, Advait Bhat, Jian Guan, Wei Wu, Rada Mihalcea, Hongning Wang, Tim Althoff, Tatia M. C. Lee, Minlie Huang

Proceedings of the National Academy of Sciences (PNAS), 2026

[6]

SocialEval: Evaluating Social Intelligence of Large Language Models

Jinfeng Zhou, Yuxuan Chen, Yihan Shi, Xuanming Zhang, Leqi Lei, Yi Feng, Zexuan Xiong, Miao Yan, Xunzhi Wang, Yaru Cao, Jianing Yin, Shuai Wang, Quanyu Dai, Zhenhua Dong, Hongning Wang, Minlie Huang

Annual Meeting of the Association for Computational Linguistics (ACL), 2025

[7]

EvoCodeBench: An Evolving Code Generation Benchmark with Domain-Specific Evaluations

Jia Li, Ge Li, Xuanming Zhang, Yunfei Zhao, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li

Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2024

[8]

DevEval: A Manually-Annotated Code Generation Benchmark Aligned with Real-World Code Repositories

Jia Li, Ge Li, Yunfei Zhao, Yongmin Li, Huanyu Liu, Hao Zhu, Lecheng Wang, Kaibo Liu, Zheng Fang, Lanshen Wang, Jiazheng Ding, Xuanming Zhang, Yuqi Zhu, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li, Bin Gu, Mengfei Yang

Findings of the Association for Computational Linguistics (ACL), 2024

E. RESEARCH GRANT AND FLAGSHIP PROGRAMS

[1]

MarioLM: Interactive World Modeling for Human-Centered AI

Xuanming Zhang, Yuxuan Chen

INTERACTIVITY RESEARCH GRANT Thinking Machines Lab

Unified evaluation–optimization architecture for interactive world modeling, faithful user simulation, and persistent experiential worlds.

[2]

Latent-RL: Internalizing Theory-of-Mind Reasoning via Mental-State Reinforcement Learning

Xuanming Zhang, Yuxuan Chen

FLAGSHIP RESEARCH PROGRAM Manuscript in preparation

Distills external multi-agent metacognition into process-supervised, efficient single-model social reasoning.

F. CURRENT MANUSCRIPTS

[1]

Are the Financial Reasoning from LLMs Credible? A Real World Test over Long-Horizon Statements

Xinke Tong, **Xuanming Zhang**[†], Tianyi Tang, An Yang, Jiayu Hu, Guojie Lin, Zhenzhen Shi, Lingfeng Zeng, Boyu Yang, Ze Xu, Hu Wei, Lin Qu, Dayiheng Liu

Under review, *EMNLP 2026* [†]Corresponding author

68,307 samples from 829 companies, uncropped statements up to 32K tokens, and structured Single-/Table-Index reasoning.

[2]

Generalization or Memorization: Evaluating Data Contamination for Large Language Models

Yihong Dong, Xue Jiang, **Xuanming Zhang**, Huanyu Liu, Zhi Jin, Bin Gu, Mengfei Yang, Ge Li

Revision in progress, *ACM Transactions on Software Engineering and Methodology*

Black-box contamination detection and trustworthy evaluation from sampled output distributions.

G. EDUCATION

Jun 2025 – Jan 2026	Stanford University GPA 4.21 / 4.00 (A+)	Visiting Undergraduate in Computer Science
Jan 2025 – Jul 2026	University of Wisconsin–Madison Dean's List	B.S. in Information Science
Sep 2021 – Dec 2024	National Elite Institute of Engineering Outstanding Student - Peking University, Jul 2023 – Dec 2024. - XbotPark, Sep 2021 – Jul 2023.	B.S. in Artificial Intelligence, transferred

H. PROFESSIONAL SERVICE

Reviewer / PC	NeurIPS (Top Reviewer , 2025), ICLR, ACL, OOPSLA, AAAI, TMLR, IEEE TII.
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I. INVITED TALKS

Jul 2026	World Artificial Intelligence Conference (WAIC) – Invited Young Distinguished Guest; sole undergraduate invitee.
Mar 2026	Fudan University.
Dec 2025	ByteDance Global Ad Recommendation Team.

J. SELECTED HONORS, RECOGNITION AND CREDENTIALS

Professional recognition	Research Scientist invitation, Anthropic Interpretability (2026); advisor to a Y Combinator-backed AI startup (2025), contributing model and product strategy.
Academic honors	Excellence Award , National College Student Innovation and Entrepreneurship Program, Ministry of Education of China (\$100,000 project funding, 2023); First-Class Scholarship , EIE (ranked 1st in comprehensive assessment); Principal's Scholarship , BNDS (highest school-wide scholarship).
Competitions	First Prize , Chinese College Students Mathematics Competition (2023); Second Prize , Chinese Physics Olympiad (2020).
Professional credential	National Level II Psychological Counselor, PCIA (2025).

K. LEADERSHIP, LANGUAGES AND INTELLECTUAL INTERESTS

Leadership	President of the Student Unions at BNDS and EIE; mentored researchers who later joined Tsinghua University, Qwen, University of Maryland, MiniMax, and other institutions.
Languages	Mandarin (native); English (professional proficiency); German (basic).
Reading & writing	Avid reader and writer in philosophy, poetry, and science fiction; two essays published in the <i>Beijing Evening News</i> .